

Model and Data Dual-Driven Double-Point Observation Network for Ultra-Short MI EEG Classification

Xu Niu , Na Lu , Ruofan Yan , and Huan Luo 

I. INTRODUCTION

Abstract—Although deep networks have succeeded in various signal classification tasks, the time sequence samples used to train the deep models are usually required to reach a certain length. Especially, in brain computer interface (BCI) research, around 3.5s-long motor imagery (MI) Electroencephalography (EEG) samples are needed to obtain satisfactory classification performance. This time-span requirement of the training samples makes real-time MI BCI systems impossible to implement based on deep networks, which restricts the related researches within laboratory and makes practical application hard to accomplish. To address this issue, a double-point observation deep network (DoNet) is developed to classify ultra-short samples buried in noise. First, an analytical solution is developed theoretically to perform ultra-short signal classification based on double-point couples. Then, a signal-noise model is constructed to study the interference of noise on classification based on double-point couples. Based on which, an independent identical distribution condition is utilized to improve the classification accuracy in a data-driven manner. Combining the theoretical model and data-driven mechanism, DoNet can construct a steady data-distribution for the double-point couples of the samples with the same label. Therefore, the conditional probability of each double-point couple of a test sample can be obtained. With a voting strategy, the samples can be accurately classified by fusing these conditional probabilities. Meanwhile, the noise interference can be suppressed. DoNet has been evaluated on two public EEG datasets. Compared to most state-of-the-art methods, the 1s-long EEG signal classification accuracy has been improved by more than 3%.

Index Terms—Data-driven, deep network, electroencephalography, model-driven, motor imagery, ultra-short signal.

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Supplemental materials include Appendix 1 "Sum-to-product formula of trigonometric function", Appendix 2 "Design and embedding methods of FIR filters", and Appendix 3 "Detailed description of the compared methods" are available at <https://gr.xjtu.edu.cn/en/web/lvna2009>.

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BRAIN computer interface (BCI) based on motor imagery (MI) electroencephalography (EEG) is a widely used paradigm to help disabled patients to restore control of outside equipment [1], [2]. In raw MI EEG data, informative signals are from the pre-motor area of the brain, where changes in power compared to the resting state provide valuable information for classification. This phenomenon is known as event-related desynchronization and synchronization (ERDS) [3]. The useful signals are buried in conduction noise and artifact interferences, resulting in a low signal-to-noise ratio and making EEG classification challenging [4].

Traditional manually-engineered classification pipelines consist of preprocessing, feature extraction, and classification procedures [5], [6]. In the preprocessing procedure, baseline offset and power line interference are eliminated using band-pass filters, while spatial filters like independent component analysis (ICA) are used to suppress conduction noise and artifacts such as electrooculogram and electromyogram. Feature extraction methods such as power spectrum calculation, common spatial pattern (CSP), and their modified versions are commonly employed [7], [8]. Finally, the extracted features are used to train a classifier such as support vector machine, linear discriminant analysis, and so on.

Despite these preprocessing efforts, the preprocessed EEG data still contains residual noise. However, since useful signals exhibit stronger periodicity than random residual noise, the residual noise has minimal effect on the feature quality of sufficiently long EEG samples. For instance, the power spectrum of a digital signal can be obtained by decomposing it into sub-band components and calculating the average of the squares of all sampling values within each sub-band [6], [7]. This averaging operation effectively suppresses random noise when the digital signal is sufficiently long. According to the Parseval theorem [9], this power spectrum calculation process is equivalent to the process composed of Fourier transform and integration in the frequency domain. Furthermore, the noise-suppressing averaging operation is also inherent in the covariance matrix calculation in CSP [8]. Therefore, most research paid attention to 3.5s-long EEG classification.

However, too long samples result in time-consuming EEG collection and model training. What's worse, users of BCIs need to continuously imagine actions for 3.5 seconds to control

external equipment, resulting in lengthy system response latencies. To the best of our knowledge, no research has explored classifying around 1s-long EEG samples without involving deep learning techniques.

In recent years, end-to-end deep learning (DL) networks have outperformed manually-engineered classification pipelines [10], [11], [12], [13]. Raw EEG can be fed directly into the end-to-end networks after band-pass filtering. Further noise reduction and feature extraction are implemented by hierarchical convolution structure [14], [15], [16], which are optimized simultaneously during network training. Additionally, a cropped training strategy has been widely used to focus on frequent information [10], [12], [17], as frequent information is more likely to be useful rather than random noise. Specifically, a sample was cropped into many segments by overlapped time windows. Each segment is used for training and given a predicted label. The final predicted label for the sample is determined through voting among all the segments. This strategy has enabled ShallowNet [10], with only 3 convolution layers, to outperform traditional classification pipelines.

Based on the cropped segment studies, some researchers have attempted to classify EEG samples shorter than 1.5 seconds. By incorporating a recurrent neural network (RNN) layer into ShallowNet, Freer et al. [18] leveraged the temporal information respectively on signals of 0.25s, 0.5s, 1s, 1.5s and 2s for classification. In several other studies, an increasing number of network layers have been stacked to improve the network performance on short EEG signal. A 7-layer convolution network and manual wavelet-based denoising were combined in [19] to implement 1s EEG classification. Zhao et al. [20] retransformed raw EEG data of 1.25s to a 3-dimension representation akin to videos and devised a 3-branch network, with each branch comprised of 3 convolution layers and 3 fully connected (FC) layers, to extract spatial-temporal features of the representations. This network was further expanded to 17 convolution or FC layers in [21] for 1.25s EEG classification. However, no satisfactory result has been obtained, because the cropped segments from an ultra-short sample are not adequate to give credible predictions, resulting in inaccurate voting results. Meanwhile, data-driven hierarchical structures are highly sensitive to changes in output from the lower hierarchical layer. Inadequate information extracted from ultra-short signals by the lower hierarchical layer makes high-level feature fusion challenging. Simply deepening networks addresses the symptoms rather than the underlying issue.

To solve ultra-short signal classification problem, a novel double-point observation network (DoNet) is proposed in this paper, which combines the advantage of analytical solutions and deep networks. Based on a rigorous analytical solution, DoNet can extract efficient features just from a pair of sampling points (double-point couple) of signal. As a substitute for a cropped segment, each double-point couple is used for network model training and gives a predicted label for voting. In other words, DoNet applies a modified cropped training strategy that takes into account the characteristics of ultra-short signals. As a result, even a very short sample also can be “cropped” into a large number of double-point couples which can provide sufficient training samples. The data-driven mechanism based on

double-point couples can facilitate DoNet to exploit useful information from raw signals buried in noise. Extensive classification experiments on 1s-long samples have been conducted on two public EEG datasets. Compared to the existing studies, 1s-long can be viewed as ultra-short for EEG signals. The results indicated that DoNet outperforms the state-of-the-art methods.

In summary, there are four main contributions:

- 1) An analytical solution is developed to utilize the information of double-point couples of single-frequency signals for classification. By applying a band-pass filter bank for spectral decomposition, the solution is generalized to broadband signals.
- 2) A signal-noise model is constructed to study the interference of noise on double-point couple based classification. Based on the model, an independent identical distribution condition is utilized to improve the classification accuracy.
- 3) The principle of introducing the data-driven mechanism to reduce noise interference is demonstrated. A scheme combining model and data dual-driven is proposed.
- 4) DoNet is designed based on the above theoretic analysis to classify multi-class multi-channel EEG signals. The classification performance on 1s-long EEG samples is superior to the state-of-the-art methods.

II. MODEL AND DATA DUAL-DRIVEN DONET

In Section A, a double-point couple from a single-frequency signal is proven to contain the information to regenerate the signal, which is the precondition that double-point couples can serve as a training sample. There is an analytical solution when the frequency is known. The analytical solution is extended to broadband signal in Section B. The classification process considering practical noise is demonstrated in Sections C and D. These four sections elaborated on model-driven mechanism. In Section E, the data-driven mechanism is introduced. DoNet is discussed in Section F.

A. Double-Point Recognition of Single-Frequency Signal

A single-frequency signal of known angular frequency ω can be written as $s(t) = A \cos \omega t$. A pair of randomly sampled points (p_1, p_2) with interval Δt can be denoted as

$$\begin{cases} p_1 = A \cos \omega t_0 \\ p_2 = A \cos \omega(t_0 + \Delta t) \end{cases} \quad (1)$$

Based on (1), the analytical solution of the signal amplitude can be solved as (the detailed procedure is in Appendix 1)

$$A = \sqrt{\left(\frac{p_1 + p_2}{2 \cos \frac{\omega \Delta t}{2}}\right)^2 + \left(\frac{p_1 - p_2}{2 \sin \frac{\omega \Delta t}{2}}\right)^2} \quad (2)$$

According to (2), the amplitude A can be calculated by any double-point couple with any given time interval Δt .

Therefore, the double-point couples from a single-frequency signal contain sufficient information to represent the signal. Based on double-point couples, the signal can theoretically be identified instead of using a long-time sequence.

There are two special cases that need to be discussed:

- 1) When $|p_1| = |p_2|$ with $\omega\Delta t = k \cdot \pi$ ($k \in \mathbb{Z}$). A cannot be calculated directly.
- 2) When

$$\omega\Delta t = k \cdot \pi + \pi/2, \quad (3)$$

there exists

$$A = \sqrt{p_2^2 + p_1^2}, \quad (4)$$

which can be calculated conveniently.

B. Double-Point Recognition Generalized to Broadband Signal

For a broadband signal $\varphi(t)$, a band-pass filter bank can be applied to decompose it into multiple sub-band components of known center frequency. Take a single-channel EEG MI signal for example, only the power within 2-32Hz needs to be considered which is demonstrated to be informative for MI classification in neuroscience. $\varphi(t)$ can be decomposed into F_1 sub-band components by a band-pass filter bank covering 2-32Hz. f_i^c ($i = 1, 2, \dots, F_1$) represents the center frequency of the band-pass filters. Suppose $\omega = 2\pi f_i^c$ and take it into (2), the amplitude of each sub-band component can be estimated accordingly. The estimated amplitudes of all the sub-band components can form the spectrum of $\varphi(t)$, based on which $\varphi(t)$ can be recognized.

A sub-band component may contain multiple single-frequency signals. In order to reduce the estimation error by making each sub-band component a quasi-single frequency signal, a filter bank design rule is proposed.

Rule 1 The pass-band of any band-pass filter should be less than one-tenth of its central frequency [22].

A rule for Δt setting is proposed to facilitate the amplitude calculation.

Rule 2 For any f_i^c , let $2\pi f_i^c \Delta t = i \cdot \pi + \pi/2$ to satisfy (3).

C. Suppress Noise Interference Via Voting of Double-Point Couples

When noise is introduced into signal $s(t)$, the amplitude A cannot be accurately calculated based on (2). The double-point couples from a signal sequence contain the same useful information but different noise due to noise randomness. In order to suppress the noise, a voting strategy is used for the following classification. Each double-point couple can be predicted as a soft label and the final predicted label of the signal is obtained by voting among all the double-point soft labels. This cropped-voting training strategy can greatly suppress the influence of random noise. Usually, the classification accuracy of the signal sequence can be increased by improving the classification accuracy of the double-point couples.

To formally describe the classification process with double-point couples, a signal-noise model and a simple 2-class classification task are constructed on a sub-band component.

1) Signal-Noise Model: Let $s^{(i)}(t)$ represent a useful sub-band component with center angular frequency ω_c , which is extracted from broadband signal-channel EEG signal $\varphi^{(i)}(t)$ ($i = 1$ or 2) with true label y_i in a binary classification scenario. $s^{(i)}(t)$ contains the information to recognize $\varphi^{(i)}(t)$.

Based on the contribution to MI classification, $\varphi^{(i)}(t)$ can be decomposed into a useful component and a generalized noise component. The noise component, useless for classification, is formed of the mixture of environmental noise and the uninformative brain signal emanating from the cerebral cortex except the sensorimotor area. The noise part can be modeled as a Gaussian signal. Compared to the fast-changing Gaussian noise, the spectrum of the useful component can be regarded as steady during an ultra-short time span. Therefore, by combining **Rule 1**, we have

$$s^{(i)}(t) = A_i \cos \omega_c t + n_i^G(t) \quad (5)$$

where $n_i^G(t)$ represents narrow-band Gaussian noise, which is obtained by band-pass filtering the Gaussian noise.

The envelope $E_i(t)$ of $n_i^G(t)$ obeys Rayleigh distribution [23]. Therefore, there exists

$$s^{(i)}(t) = A_i \cos \omega_c t + E_i(t) \cos(\omega_c t + \theta_i), \quad (6)$$

where θ_i is the constant phase difference in each $s^{(i)}(t)$. Based on the phasor method, we have

$$\begin{cases} s^{(i)}(t) = A'_i(t) \cos[\omega_c t + \theta'_i(t)] \\ A'_i(t) = \sqrt{A_i^2 + E_i^2(t) + 2A_i E_i(t) \cos \theta_i} \\ \theta'_i(t) = \tan^{-1} \frac{E_i(t) \sin \theta_i}{A_i + E_i(t) \cos \theta_i} \end{cases}, \quad (7)$$

where $[A'_i(t)]_t^T = 0$ obeys Rician distribution $R(A_i, \sigma_i)$ [23], whose probability density function is

$$f_i(x) = \frac{x^2}{\sigma_i^2} e^{-\frac{(x^2 - A_i^2)}{2\sigma_i^2}} I_0\left(\frac{x A_i}{\sigma_i^2}\right), \quad x > A_i, \quad (8)$$

where $I_0(\cdot)$ is the modified Bessel function of the first kind with order zero.

The SNR (signal-to-noise ratio) of EEG signals is very low. Thus, we propose

Assumption 1: $\theta'_i(t) \approx \theta_i$ because $E_i(t) \gg A_i$ for any t .

2) Classification By Analytical Solution: A double-point dataset $\{\mathbf{p}_i^n\}_{n=1}^N$ is sampled from $s^{(i)}(t)$ with a fixed time interval Δt . $\mathbf{p}_i^n = (p_{i,1}^n, p_{i,2}^n)$ and

$$\begin{cases} p_{i,1}^n = A'_i(t_n) \cos[\omega_c t_n + \theta'_i(t_n)] \\ p_{i,2}^n = A'_i(t_n + \Delta t) \cos[\omega_c(t_n + \Delta t) + \theta'_i(t_n + \Delta t)] \end{cases}. \quad (9)$$

A binary classification task is set to classify $\{\mathbf{p}_1^n\}_{n=1}^N$ and $\{\mathbf{p}_2^n\}_{n=1}^N$ which are from $s^{(1)}(t)$ and $s^{(2)}(t)$ respectively.

The amplitude $A'_i(t_n)$ of $(p_{i,1}^n, p_{i,2}^n)$ can be estimated as $\tilde{A}_i^n$ based on (2). When $\Delta t \rightarrow 0$, $A'_i(t_n) = A'_i(t_n + \Delta t)$. Thus $\tilde{A}_i^n = A'_i(t_n)$ and

$$[\tilde{A}_i^n]_{n=1}^N \sim R(A_i, \sigma_i). \quad (10)$$

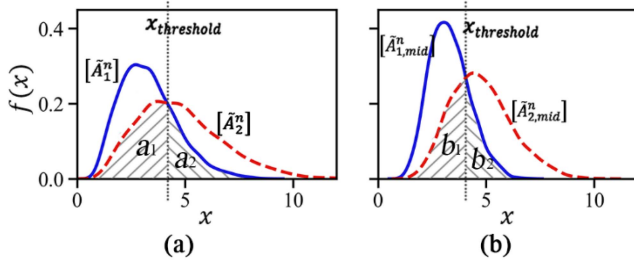


Fig. 1. Probability density function of simulated Rician distribution. (a) Blue line denotes the probability density function $f(\tilde{A}_1^n)$ of $[\tilde{A}_1^n]_{n=1}^N$ and red dashed line denotes $f(\tilde{A}_2^n)$. (b) The variances of $f(\tilde{A}_{1,mid}^n)$ and $f(\tilde{A}_{2,mid}^n)$ are the smaller than those of $f(\tilde{A}_1^n)$ and $f(\tilde{A}_2^n)$, which will be used for demonstration in Section II-D.

The expectation and variance are

$$\begin{cases} E[\tilde{A}_i^n] = E[R(A_i, \sigma_i)] \\ D[\tilde{A}_i^n] = D[R(A_i, \sigma_i)] \end{cases} \quad (11)$$

Based on the probability density curves of $[\tilde{A}_1^n]_{n=1}^N$ and $[\tilde{A}_2^n]_{n=1}^N$ as shown in Fig. 1(a), if $\tilde{A}_i^n < x_{threshold}$, the double-point couple is predicted as label y_1 . Otherwise, it is predicted as y_2 . According to Fig. 1, the classification accuracy can be calculated as

$$Acc = \frac{(1 - a_2) + (1 - a_1)}{2} = 1 - \frac{a_{overlap}}{2}, \quad (12)$$

where a_1 and a_2 are the mistakenly classified ratios of the two classes. To sum up, the classification accuracy relies on the ratio of the overlap area $a_{overlap}$ between the probability density functions $f(\tilde{A}_1^n)$ and $f(\tilde{A}_2^n)$.

D. Independent Identical Distribution Condition and Hilbert Envelope Based Mid-Value Method

1) *Improve Classification Accuracy Based on Independent Identical Distribution Condition*: The setting of $\Delta t \rightarrow 0$ is not practical in application. Suppose t_{max} and t_{min} are the lower and upper bounds for Δt . When $t_{min} \leq \Delta t \leq t_{max}$, $[A_i'(t_n)]_{n=0}^N$ and $[A_i'(t_n + \Delta t)]_{n=0}^N$ are independent and identically distributed.

Specifically, $[A_i'(t_n)]_{n=0}^N \sim R_1(\sigma_i, A_i)$, $[A_i'(t_n + \Delta t)]_{n=0}^N \sim R_2(\sigma_i, A_i)$. $R_1(\sigma_i, A_i)$ and $R_2(\sigma_i, A_i)$ have the same parameters with $R(\sigma_i, A_i)$ but they are independent. At the same time, Δt is set based on **Rule 2**. Based on (4), the estimated amplitude $\tilde{A}_{i,mid}^n$ of $(p_{i,1}^n, p_{i,2}^n)$ can be obtained as

$$\tilde{A}_{i,mid}^n = \sqrt{(p_{i,1}^n)^2 + (p_{i,2}^n)^2}. \quad (13)$$

As illustrated in (12), the classification accuracy based on $\tilde{A}_{i,mid}^n$ relies on the overlap area $a_{overlap}^{mid}$ between the probability density functions $f(\tilde{A}_{1,mid}^n)$ and $f(\tilde{A}_{2,mid}^n)$. If $a_{overlap}^{mid}$ is less than $a_{overlap}$, the classification accuracy will be improved. Therefore, we formulate (14) to calculate the expectation and

variance of $[\tilde{A}_{i,mid}^n]_{n=1}^N$.

$$\tilde{A}_{i,mid}^n = k_i^n A_i'(t_n) + (1 - k_i^n) A_i'(t_n + \Delta t), \quad 0 < k_i^n < 1, \quad (14)$$

where k_i^n is the combination weight, meaning $\tilde{A}_{i,mid}^n$ locates between $A_i'(t_n)$ and $A_i'(t_n + \Delta t)$. Considering the independent identical distribution condition, it can be obtained

$$\begin{cases} E[\tilde{A}_{i,mid}^n] = E[R(A_i, \sigma_i)] \\ D[\tilde{A}_{i,mid}^n] = D[k_i^n A_i'(t_n)] + D[(1 - k_i^n) A_i'(t_n + \Delta t)] \end{cases} \quad (15)$$

k_i^n varies with n , but there exists the supremum D_{upper} of $D[\tilde{A}_{i,mid}^n]$. Suppose

$$\begin{aligned} D_{upper} &= D[k_i A_i'(t_n)] + D[(1 - k_i) A_i'(t_n + \Delta t)] \\ &= k_i^2 D[A_i'(t_n)] + (1 - k_i)^2 D[A_i'(t_n + \Delta t)] \\ &= (1 + 2k_i^2 - 2k_i) D[R(A_i, \sigma_i)]. \end{aligned} \quad (16)$$

With fixed $D[R(A_i, \sigma_i)]$, D_{upper} approaches the minimum when $k_i \rightarrow 0.5$. On the contrary, D_{upper} approaches the maximum when k_i equals $\text{Max}\{[k_i^n]_{n=1}^N\}$ or $\text{Min}\{[k_i^n]_{n=1}^N\}$.

Since $0 < \text{Max}\{[k_i^n]_{n=1}^N\}$, $\text{Min}\{[k_i^n]_{n=1}^N\} < 1$, we have

$$1 + 2k_i \cdot k_i - 2k_i < 1. \quad (17)$$

Therefore,

$$D[\tilde{A}_{i,mid}^n] \leq D_{upper} < D[R(A_i, \sigma_i)]. \quad (18)$$

Compared to the case of $\Delta t \rightarrow 0$, $E[\tilde{A}_{i,mid}^n] = E[\tilde{A}_i^n]$ but $D[\tilde{A}_{i,mid}^n] < D[\tilde{A}_i^n]$. Since the variances decrease when the expectations are unchanged, the overlap area will decrease as shown in Fig. 1(b). Accordingly, the classification accuracy based on double-point couples can be improved.

2) *Variance Adjustment and Hilbert Envelope Based Mid-Value Method*: Based on the above analytical model, the probability density functions of two signal sequences can be modified manually with double point couple calculation and used to improve classification accuracy. Alternatively, the envelope value $A_i'(t)$ can be calculated using the Hilbert transform [24] and then k_i^n is set as 0.5 in (14) to get the mid-value of each pair of the envelope values. In this way, new amplitude datasets $[\tilde{A}_1(t_n)]_{n=0}^N$ and $[\tilde{A}_2(t_n)]_{n=0}^N$ are obtained, whose probability density functions have reduced overlap area via pure variance adjustment. This method is named as Hilbert envelope based mid-value method (HEM) in this paper.

However, an even smaller overlap area than the above result may be obtained by adding expectation adjustment. Since it is difficult to manually adjust expectations, a data-driven mechanism is needed, which leads to our solution.

To sum up, two inductions can be obtained to improve the classification accuracy.

Induction 1 Δt needs to be set large enough to satisfy the independent identical distribution condition.

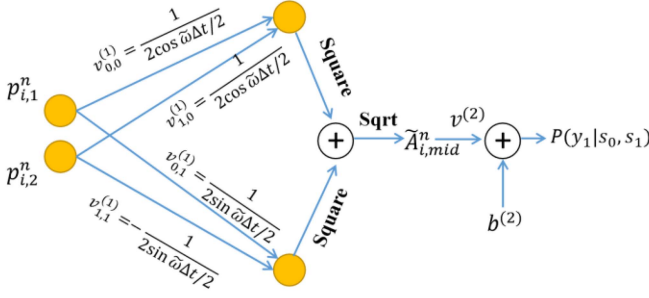


Fig. 2. SfNet. Square denotes the quadratic calculation and Sqrt denotes the square root calculation.

Induction 2 The combination weight $[k_i^n]_{n=1}^N$ can be adaptively adjusted based on a data-driven neural network. Besides variance adjustment, the expectations $E[\tilde{A}_1^n]$ and $E[\tilde{A}_2^n]$ would also be tuned to move further away from each other, which leads to a smaller overlap area.

The signal-noise model and the inductions will be verified on real-world data in Sections IV-B and IV-C.

E. Data-Driven Mechanism

Model-driven signal classification mechanism has been discussed above. Based on **Induction 2**, model-driven and data-driven mechanisms can be combined.

To implement (2), a simple model and data dual-driven network working on single-frequency signal (termed as SfNet) is constructed as shown in Fig. 2. The network consists of two full-connected (FC) layers. $v_{0,0}^{(1)}, v_{1,0}^{(1)}, v_{0,1}^{(1)}, v_{1,1}^{(1)}$ denote the weight parameters of the first FC layer, which can be initialized as the coefficients of the p_1, p_2 in (2). With quadratic sum and square root operations, the estimated amplitude $\tilde{A}_{i,mid}^n$ can be obtained as

$$\tilde{A}_{i,mid}^n = \sqrt{\left[v_{0,0}^{(1)}p_{i,1}^n + v_{1,0}^{(1)}p_{i,2}^n\right]^2 + \left[v_{0,1}^{(1)}p_{i,1}^n + v_{1,1}^{(1)}p_{i,2}^n\right]^2}. \quad (19)$$

The second FC layer aims to implement classification based on $\tilde{A}_{i,mid}^n$, whose parameters are $v^{(2)}, b^{(2)}$. SfNet substitutes the analytical model to classify $\{p_i^n\}_{n=1}^N$. Specifically, $\{p_i^n\}_{n=1}^N$ is divided into a training set and a test set. By minimizing the training loss, SfNet can solve the classification task end-to-end. During the training step, SfNet can learn to flexibly adjust $[k_i^n]_{n=1}^N$. In this way, the signal amplitude can be estimated with data by neural network constructed via analytical model.

The advantages of introducing the data-driven mechanism:

- 1) The signal frequency need not be known accurately.
- 2) SfNet can still work when **Assumption 1** is not satisfied.
- 3) A smaller overlap area can be obtained as in Fig. 1 following **Induction 2**. Accordingly, higher accuracy can be achieved.

In practice, the amplitude features of the various sub-band components need to be fused for recognition. The information in multiple channels of EEG also needs to be fused. Obviously,

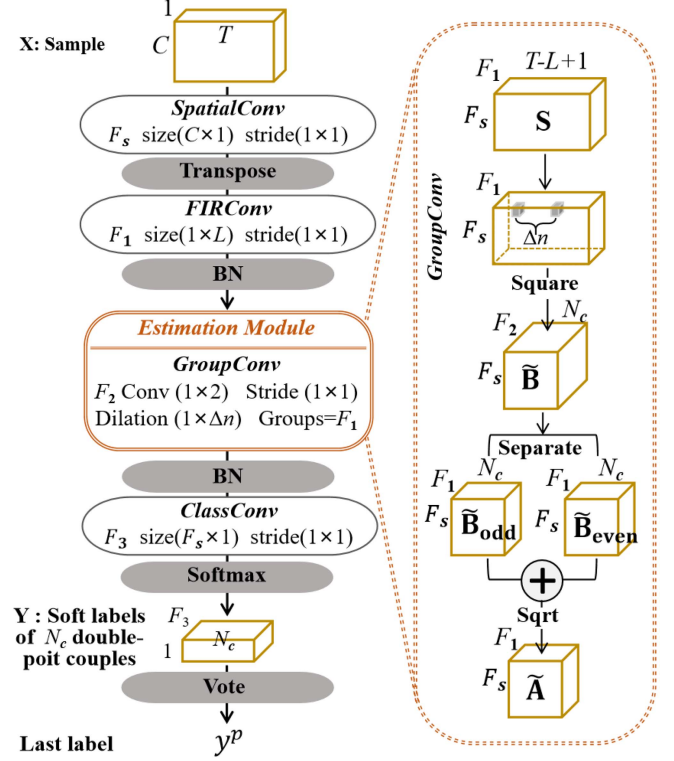


Fig. 3. Naive-DoNet consists of SpatialConv, FIRConv, an estimation module, ClassifierConv, and a voting module. Transpose denotes matrix-axis transformation. BN denotes batch normalization. Square denotes square activation.

not all the sub-band components are informative. It is very time consuming to manually select and fuse the extracted features. To address these issues conveniently, DoNet is developed to implement multi-channel EEG classification in an end-to-end manner.

F. Double-Point Observation Network

The structure of Do-Net (Double-point observation network) is illustrated in Fig. 3, which is an extension of SfNet. SpatialConv aims to reduce artifact interference such as electrooculography (EOG) and electromyography (EMG) in raw EEG by implementing spatial filtering. FIRConv serves as the band-pass filter bank mentioned in Section II-B. The estimation module and ClassifierConv implement parallel arrangement and fusion of multiple SfNets.

1) Network Structure: An EEG signal with C channels and time-length T is fed into DoNet as a sample. The SpatialConv output is of size $(F_s \times 1 \times T)$, which represents F_s spatial-filtered signals (termed as SF signal). Its shape is transformed as $(1 \times F_s \times T)$ by transpose operation to be fed into FIRConv.

FIRConv is a temporal convolution layer, whose kernels are set as F_1 FIR (finite impact response) band-pass filters [12]. L represents the length of the filter. By convolution calculation between a SF signal and a FIR band-pass filter (FIRConv kernel), a sub-band component of the SF signal can be obtained (see Appendix 2 for details). Thus, the output of FIRConv $S \in R^{F_1 \times F_s \times (T-L+1)}$ contains $F_1 \times F_s$ sub-band components,

which are of the same nature as $s^{(i)}(t)$ of Section II-C. It should be noted that FIRConv is not updated during network training.

The GroupConv of the estimation module is used to sample double-point couples from the $F_1 \times F_s$ components. The double-point sampling interval is set as a dilation rate $\Delta n = f_s \times \Delta t$, where f_s is the EEG signal sampling frequency and Δt is the time interval in the above discussed theory. Thus $N_c (= T - L + 1 - \Delta n)$ groups of double-point couples are obtained, each one of which contains double-point couples of $F_1 \times F_s$. The hyper-parameter ‘‘Groups’’ of GroupConv is set as F_1 which equals the input channel number of GroupConv i.e., sub-band component number.

In this way, convolution operation can only be implemented in any single convolution channel and thus the amplitude of each sub-band component can be estimated independently. Based on (19), the kernel number F_2 is set as $2 \times F_1$. The square activation function is applied after GroupConv. In this way, a double-point couple of each sub-band component can be transformed to two quadratic terms, which form $\tilde{\mathbf{B}} \in R^{2F_1 \times F_s \times N_c}$. $\tilde{\mathbf{B}}$ can be separated as

$$\tilde{\mathbf{B}}_{odd} = \tilde{\mathbf{B}}_{2i-1,:} \text{ and } \tilde{\mathbf{B}}_{even} = \tilde{\mathbf{B}}_{2i,:}, i = 1, 2, \dots, F_1. \quad (20)$$

The estimated amplitude matrix can be written as

$$\tilde{\mathbf{A}} = \sqrt{\tilde{\mathbf{B}}_{odd} + \tilde{\mathbf{B}}_{even}}, \quad (21)$$

which is composed of the estimated amplitudes of all the double-point couples of one sample. The detailed algorithm pseudocode is given in Algorithm 1. Each group of double-point couples can be given a predicted soft label by ClassifierConv. The ClassifierConv output $\mathbf{Y} \in R^{F_3 \times N_c}$ contains the soft labels of the N_c groups of double-point couples.

In short, the combination of the estimation module and ClassifierConv is equivalent to the parallel connection of $F_1 \times F_s$ SfNets. Each SfNet independently works on each one of $F_1 \times F_s$ sub-band components.

2) Training and Test Strategy: In the training step, a cross-entropy value is calculated between the sample’s true one-hot label $\mathbf{y}^t \in R^{F_3}$ and the soft label of each group. As shown in (22), the average of the cross-entropy values is used as the training loss to update DoNet parameters.

$$Loss = -\frac{1}{N_c} \sum_{j=1}^{N_c} \sum_{i=1}^{F_3} \mathbf{y}_i^t \log \mathbf{Y}_{i,j}^s. \quad (22)$$

By this way, a steady data-distribution for the double-point couples of the samples with the same label can be learned.

The soft label given to each group of double-point couples consists of the probability values that the group belongs to each category i.e., the conditional probability. In the test step, the final predicted label y^p of the sample is determined by the majority voting over all the N_c soft labels, which is obtained as

$$y^p = \text{Mo} \left[\left\{ \arg\max_i \mathbf{Y}_{i,j}^s \right\}_{j=1}^{N_c} \right], \quad (23)$$

where Mo ($\cdot$) means the majority voting operation.

Algorithm 1: Estimating Amplitudes Via the Estimation Module.

Input: $F_1 \times F_s$ sub-band components

$\mathbf{S} \in R^{F_1 \times F_s \times (T-L+1)}$

Output: Estimated amplitudes $\tilde{\mathbf{A}} \in R^{F_1 \times F_s \times N_c}$

Define: Parameter matrix $\mathbf{W} \in R^{F_2 \times 2}$ of GroupConv

$F_2 = 2F_1$

Dilation rate Δn of GroupConv

- 1: # i : index of F_s spatial-filtered signals; j : index of F_1 sub-band components of each spatial-filtered signal; n : index of the collected double-point couple
 - 2: **for** $i \leftarrow 1$ to F_s **do**
 - 3: **for** $j \leftarrow 1$ to F_1 **do**
 - 4: **for** $n \leftarrow 1$ to N_c **do**
 - 5: # Collect the n -th double-point couple (p_1, p_2) of the j -th sub-band component of the i -th spatial-filtered signal
 - 6: $p_1 = \mathbf{S}_{j,i,n}$ $p_2 = \mathbf{S}_{j,i,n+\Delta n}$
 - 7: # Estimate the amplitude of the double-point couple
 - 8: $\tilde{\mathbf{A}}_{j,i,n} = \frac{1}{\sqrt{[\mathbf{W}_{2j-1,1}p_1 + \mathbf{W}_{2j-1,2}p_2]^2 + [\mathbf{W}_{2j,1}p_1 + \mathbf{W}_{2j,2}p_2]^2}}$
 - 9: **End for**
 - 10: **End for**
 - 11: **End for**
-

In this way, the common information of all the double-point couples from one sample can be aggregated and the influence of the stochastic noise can be suppressed.

In addition, two batch normalization (BN) operations are added to accelerate training convergence.

3) Setting of Hype-Parameter L: For FIR band-pass filter, the larger its length L is, the narrower its pass-band can be set. Based on Rule 1, L should be set as large as possible. However, the FIRConv output length $T-L+1$ will be too small if L is set too large. The number of the captured double-point couple groups becomes too few, resulting in poor data-driven performance. Therefore, the optimal L needs to be obtained by pre-experiments.

III. EXPERIMENT SPECIFICATION

A. Sample-Set Description

BCI Competition IV-2a [25] and Motor Movement/Imagery Database (termed as MMIDB) were used for experiments.

BCIC IV-2a includes two sessions of EEG data from nine subjects. In each session, each subject was stimulated by a cue to imagine one of the four-category motor imageries (left-hand, right-hand, feet, and tongue). EEG data were collected via 22 electrodes and sampled at 250 Hz. The first session was used in our experiments. The EEG signal of 1s length after the cue was cropped as one sample. It was filtered between 1-48Hz to eliminate direct-current offset and power line interference. Meanwhile, it was multiplied by 10^6 to avoid gradient diffusion.

Finally, a balanced four-category sample set consisting of 288 samples of size (22, 250) was built for each subject.

MMIDB was collected via 64 electrodes with a sampling rate of 160 Hz [26], [27]. The first 10 subjects of the 109 subjects were selected for our experiments. For each subject, there are 90 EEG trials on MI tasks including moving the left hand, right hand, both feet, and both hands. A four-category sample set consisting of 90 samples of size (64, 160) was built for each subject.

B. Experiment Procedures

Extensive experiments have been conducted to verify the network performance, the signal-noise model, **Inductions 1** and **2**.

By some pre-experiments, the hyper-parameters were selected. For BCIC IV-2a, F_s , F_1 , F_2 and F_3 were set as 16, 24, 48 and 4 respectively. Based on **Rule 2**, the center frequency of the 24 FIR band-pass filters (FIRConv kernels) was respectively set as $(1.875 + i \times 1.25)\text{Hz}$ where $i = 1, 2, \dots, 24$. The FIR filters' length L was set as 51. Thus the output of FIRConv **S** is of size $(24 \times 16 \times 200)$. Based on **Induction 1**, Δt was set as 0.4s and $\Delta n = 250\text{Hz} \times 0.4\text{s} = 100$. In this way, 100 groups of double-point couples can be captured from each sample.

DoNet-100 (DoNet of $\Delta n = 100$) was evaluated by 4-fold cross-validation, which was run 10 times and the training procedure was iterated for 600 epochs with a batch size of 8. The learning rate is 10^{-3} and Adam with default parameters serves as an optimizer. The highest classification accuracy among 10 runs was recorded.

For MMIBD, $\Delta n = 160\text{Hz} \times 0.4\text{s} = 64$, and L was changes to 33 based on the ratio of sampling rates for these two datasets.

Various state-of-the-art methods have been incorporated for comparison, including ShallowNet [10], ShallowNet with cropped training strategy [10], EEGNet [11], KFCNet [12], MFBCNN [13], EEG-TCNet [14], C-LSTM [18], 3-D CNN [20], LMDA-Net [15] and ATCNet [17]. (Please see Appendix 3 for the detailed description of the baseline networks)

C. Visualization Procedures to Verify the Signal-Noise Model and Induction 2

As shown in Fig. 3, the FIRConv output **S** and the estimated amplitude matrix **A** of a trained DoNet-100 can be pulled out for visualization and analysis. $\mathbf{S} \in R^{F_1 \times F_s \times (T-L+1)}$ consists of F_s SF signals, each one of which has been decomposed into F_1 sub-band components.

To verify the signal-noise model, let $[A_{left}^{i,j}(n)]_{n=0}^N$ ($N = T - L + 1$) represents the envelope of $\mathbf{S}_{i,j}$, (i.e., the i -th sub-band component of the j -th SF signal) of a left-hand MI sample, which can be calculated based on Hilbert transform [24]. $[A_{left}^{i,j}(n)]_{n=0}^N$ corresponds to $[A'_i(t)]_{t=0}^T$ of the signal-noise model. $[A_{left}^{i,j}(n)]_{n=0}^{N \times N_{all}}$ was formed by the envelopes of the sub-band component of all the left-hand MI samples, whose distribution is named as left-hand intrinsic envelope-distribution of $\mathbf{S}_{i,j}$. The probability density function $f([A_{left}^{i,j}(n)])$ of the intrinsic envelope-distribution was plotted for comparison with the Rician distribution. In the same way, $f([A_{right}^{i,j}(n)])$ was

plotted. The overlapped area between the probability density curves was calculated.

To verify **Inductions 1** and **2**, the estimated amplitude set $[\tilde{A}_{100, left}^{i,j}(n)]_{n=0}^{N_c}$ was taken out from a left-hand sample, where $\tilde{A}_{100, left}^{i,j}(n)$ is equal to $\tilde{A}_{i,j,n}$ in the DoNet-100. $[\tilde{A}_{100, left}^{i,j}(n)]_{n=0}^{N_c \times N_{all}}$ of all the left-hand samples was recorded. Thus, $f([A_{left}^{i,j}(n)])$ and $f([A_{right}^{i,j}(n)])$ have been estimated as $f([A_{100, left}^{i,j}])$ and $f([\tilde{A}_{100, right}^{i,j}])$ by DoNet-100. The overlapped area variation between them was shown.

To further verify **Induction 2** by comparing DoNet with HEM, $[\bar{A}_{left}^{i,j}(n)]_{n=0}^{N_c}$ and $[\bar{A}_{right}^{i,j}(n)]_{n=0}^{N_c}$ are obtained based on the Hilbert envelope based mid-value method (HEM). Specifically, $\bar{A}_{left}^{i,j}(n) = \frac{\bar{A}_{left}^{i,j}(n) + \bar{A}_{left}^{i,j}(n + \Delta n)}{2}$ where $\Delta n = 100$. The overlapped area of $f([\bar{A}_{left}^{i,j}(n)])$ and $f([\bar{A}_{right}^{i,j}(n)])$ is calculated.

In addition, whether DoNet-1 ($\Delta t \rightarrow 0$ in the signal-noise model) could estimate amplitudes accurately should also be verified. Keep the trained SpatialConv and FIRConv unchanged, and the trained DoNet-100 was re-trained after Δn is modified as 1. In this way, a trained DoNet-1 which works on the same **S** as the trained DoNet-100 was obtained. Thus, $f([A_{left}^{i,j}(n)])$ and $f([A_{right}^{i,j}(n)])$ have been estimated as $f([\tilde{A}_{1, left}^{i,j}])$ and $f([\tilde{A}_{1, right}^{i,j}])$ by DoNet-1. The overlapped area variation between them was shown.

The above visualization procedures can be repeated on each sub-band component of each SF signal. The feet and tongue MI samples can also be analyzed based on the above procedures.

IV. RESULTS AND VISUALIZATION

A. Performance Comparison

The performance of DoNet and the compared methods on BCIC IV-2a are shown in Table I. The results with the highest accuracy are highlighted in bold. It can be seen that DoNet significantly outperformed the other methods. DoNet obtained the best performance on seven out of nine subjects. The highest average accuracy was also obtained by DoNet. As shown in Table II, DoNet also outperformed the other methods on MMIDB. The performance of DoNet has been verified.

Furthermore, we implemented ANOVA and the obtained p -values are reported in Table III. Non-significant results ($p > 0.05$) are highlighted in bold. The analysis reveals that the accuracy improvement based on DoNet is significant, with only a few instances of non-significant differences observed.

Lastly, Table IV compares the model complexity of the methods. Only KFCNet and EEGNet are more compact than DoNet, but the performance of DoNet improves over these two methods significantly. Specifically, compared to KFCNet, DoNet improves the average accuracy by 15.20% on BCIC IV 2a and 4.26% on MMIBD. Compared to EEGNet, DoNet increases the average accuracy by 7.32% on BCIC IV 2a and 13.21% on MMIBD.

B. Analysis on Key Hyper-Parameters

The performance comparison among DoNets with different settings of L is shown in Table V. There is no significant

TABLE I
PERFORMANCE COMPARISON OF DoNet AND THE BASELINE METHODS ON BCIC IV 2A (%)

Method	S1	S2	S3	S4	S5	S6	S7	S8	S9	Ave.
C-LSTM	61.11	55.21	60.42	55.90	55.90	52.08	56.25	56.60	76.39	58.87
ShallowNet	62.15	55.21	59.38	58.33	58.33	51.04	59.37	56.25	76.39	59.61
MFSBCNN	62.15	51.39	60.07	56.94	56.94	51.74	61.11	59.03	77.78	59.68
ShallowNet (cropped)	60.42	50.35	69.10	57.99	57.99	59.38	68.75	61.46	77.43	62.54
KFCNet	65.28	53.82	68.40	63.19	71.88	63.54	77.09	63.19	73.96	66.71
EEGNet	75.79	65.28	78.12	73.96	73.96	63.19	80.21	71.88	88.88	74.59
3-D CNN	77.40	60.14	82.93	72.29	75.84	68.99	76.04	76.86	84.67	75.02
LMDA-Net	74.31	71.18	80.21	73.96	65.28	67.71	86.81	79.17	92.01	76.74
EEG-TCNet	78.47	62.15	79.86	72.57	80.56	73.61	77.08	79.17	91.67	77.24
ATCNet	80.56	67.36	79.86	79.17	81.95	73.96	77.78	77.08	90.28	78.67
DoNet-100	80.56	64.59	86.46	74.65	85.76	78.82	91.32	81.25	93.75	81.91

TABLE II
PERFORMANCE COMPARISON OF DoNet AND THE BASELINE METHODS ON MMIDB (%)

Method	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	Ave.
EEG-TCNet	51.19	62.33	70.24	59.52	44.57	59.52	51.72	44.14	57.10	59.52	55.99
MFSBCNN	53.57	70.24	66.67	57.14	53.57	70.24	57.14	51.19	60.71	59.52	60.00
ShallowNet	54.76	75.00	69.05	60.71	60.71	73.81	61.90	59.52	67.86	60.71	64.40
EEGNet	84.52	69.05	91.67	77.38	63.10	80.95	80.95	65.48	71.43	66.67	75.12
LMDA-Net	82.14	65.48	78.57	79.76	70.24	78.57	77.38	65.48	75.00	82.14	75.48
ShallowNet (cropped)	84.52	76.19	77.38	85.71	66.67	79.76	83.33	71.43	72.62	72.62	77.02
ATCNet	85.71	77.38	82.14	88.09	61.90	88.10	89.29	70.24	77.38	75.00	79.52
C-LSTM	82.14	84.52	86.90	84.52	72.62	86.90	83.33	69.05	85.71	83.33	81.90
KFCNet	80.95	84.52	91.67	88.10	72.62	86.91	91.67	82.14	83.33	79.76	84.17
DoNet	84.52	90.48	92.86	89.29	76.19	92.86	100.0	86.90	90.48	79.76	88.33

TABLE III
ANOVA ON BCIC IV 2A AND MMIDB (P-VALUE)

BCIC IV 2a / MMIDB	DoNet-100									
	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
C-LSTM	1.3e-17 / 2.2e-05	3.5e-11 / 1.1e-5	5.5e-15 / 5.6e-10	3.3e-17 / 1.9e-8	1.3e-22 / 0.94	6.5e-20 / 6.3e-7	8.1e-26 / 8.1e-14	2.0e-15 / 8.6e-11	0.035 / 0.0018	- / 0.052
ShallowNet	1.8e-17 / 1.3e-18	1.9e-9 / 7.6e-11	1.1e-18 / 8.6e-17	1.2e-7 / 1.8e-16	9.5e-22 / 1.6e-7	1.8e-19 / 7.2e-13	1.6e-21 / 1.9e-17	2.5e-18 / 5.8e-14	0.0095 / 1.4e-11	- / 4.4e-11
MFSBCNN	1.5e-16 / 6.5e-17	1.5e-12 / 7.7e-13	3.4e-19 / 1.9e-15	1.4e-8 / 9.8e-18	9.3e-21 / 1.7e-9	2.6e-19 / 1.8e-12	7.5e-21 / 9.7e-21	8.6e-18 / 1.1e-16	0.015 / 6.8e-14	- / 7.3e-11
ShallowNet (cropped)	5.0e-18 / 0.00012	2.7e-12 / 3.1e-9	1.0e-15 / 4.0e-13	4.3e-11 / 0.00025	5.7e-19 / 0.0041	6.4e-17 / 1.3e-9	7.8e-18 / 1.6e-13	1.8e-15 / 1.1e-10	0.0092 / 1.2e-10	- / 2.4e-6
KFCNet	2.4e-17 / 7.8e-8	4.1e-10 / 8.1e-6	9.3e-17 / 0.0035	1.1e-11 / 0.012	6.7e-15 / 0.0076	1.6e-14 / 0.0016	1.1e-16 / 4.3e-7	2.4e-15 / 3.9e-5	0.0017 / 6.6e-7	- / 0.37
EEGNet	1.4e-9 / 0.0023	0.29 / 2.4e-13	3.7e-12 / 1.4e-6	0.00069 / 6.1e-10	1.0e-12 / 5.6e-5	1.3e-12 / 7.0e-9	3.5e-12 / 1.8e-15	9.6e-11 / 2.5e-12	0.70 / 9.3e-10	- / 1.1e-7
LMDA-Net	3.8e-11 / 0.00036	7.6e-9 / 4.8e-8	3.7e-6 / 2.9e-12	0.15 / 5.4e-10	4.3e-16 / 0.0019	1.6e-6 / 2.5e-12	1.3e-11 / 1.2e-12	0.0046 / 7.3e-11	0.65 / 1.3e-8	- / 1.00
EEG-TCNet	0.0024 / 1.0e-26	0.93 / 1.9e-22	2.2e-5 / 1.1e-26	0.62 / 5.5e-25	1.7e-6 / 1.5e-17	4.2e-5 / 5.5e-24	4.3e-16 / 7.5e-28	0.23 / 2.1e-21	0.81 / 1.8e-21	- / 3.2e-21
ATCNet	0.0019 / 0.00051	1.5e-5 / 3.4e-8	1.5e-9 / 1.8e-7	8.2e-7 / 0.00046	0.23 / 0.00012	2.7e-5 / 1.0e-5	4.2e-17 / 2.4e-9	0.0028 / 4.4e-11	0.74 / 3.1e-10	- / 3.6e-5

difference in the average accuracy when L is 31, 41, and 51. However, setting L to 11 or 91 leads to a significant deterioration in the network performance. This is because optimal performance is achieved when there is a balance between the model-driven and data-driven processes, which occurs within a specific range of L values. Specifically, when the filter length is too short, the filtered signal lacks narrow-band characteristics, resulting in poor performance of the model-driven process based on the analytic solution. On the contrary, when the filter

length is too long, the number of the collected double-point couples decreases significantly. Consequently, the data-driven DL network training process suffers from poor performance. In summary, the best results are achieved when there is a balance between the model-driven and data-driven processes, which are regulated by L .
The performance comparison among DoNets with different settings of Δn is shown in Table VI. It can be seen that the best performance was obtained when Δn was set as 100. Therefore,

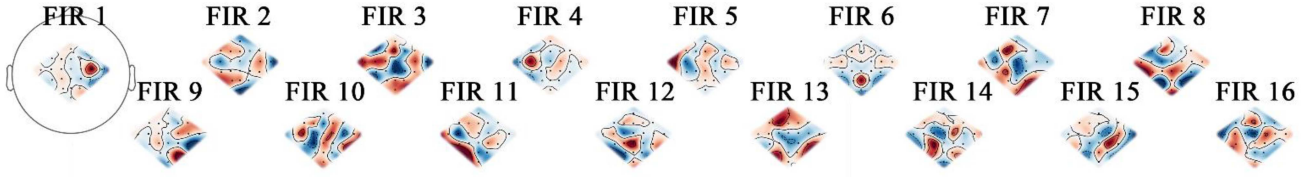


Fig. 4. Topographic maps of the spatial filters. The black dots represent the electrode locations, from where the EEG signals were collected. Each topographic map is colored by the weight assigned for each electrode by the spatial filter. Red represents positive weight and blue represents negative. The deeper the color, the greater the magnitude of the weight.

TABLE IV
MODEL COMPLEXITY COMPARISON OF DoNet AND BASELINES

Method	BCIC IV 2a		MMIDB	
	Params	Flops(k)	Params	Flops(k)
3-D CNN	3341868	34299	-	-
MFSBCNN	150684	49329	419484	143409
ATCNet	94640	19710	95984	53478
ShallowNet (cropped)	41124	13027	108324	37707
C-LSTM	39944	11294	90344	32177
ShallowNet	38604	12966	105804	37645
LMDANet	4734	17109	5112	49595
EEG-TCNet	4272	4572	4944	13140
KFCNet	2107	2849	2509	1378
EEGNet	1460	1618	2132	4629
DoNet-100	3288	4773	3528	2277

Induction 1 was verified. The meaning of this hyper-parameter will be further explained in Sections IV-C and IV-D.

C. Intrinsic Envelope-Distribution and Its Estimation

As mentioned in Section III-C, $\mathbf{S}$ and $\tilde{\mathbf{A}}$ of a trained DoNet-100 on subject 9 were used for visualization.

First, the significant SF signals which made a large contribution to classification need to be sought for better visualization. Since each SF signal was obtained by the corresponding spatial filters (SpatialConv kernel), the topographic maps of the spatial filters were plotted to select the dominant SF signals.

As shown in Fig. 4, the 1st, 4th, 6th and 12th spatial filters' topographic maps have clearer peaks. Therefore, the SF signals $\mathbf{S}_{:,1,:}$, $\mathbf{S}_{:,4,:}$, $\mathbf{S}_{:,6,:}$ and $\mathbf{S}_{:,12,:}$ obtained by the four FIR filters were selected. Among them, $\mathbf{S}_{:,1,:}$ and $\mathbf{S}_{:,4,:}$ respectively contain the high-quality information of right- and left-side premotor areas, which can be used to classify left- and right-hand MI samples [3]. $\mathbf{S}_{:,6,:}$ and $\mathbf{S}_{:,12,:}$ contain the high-quality information from the longitudinal fissure, which can be used to classify feet and tongue MI samples [3].

$f([\tilde{\mathbf{A}}_{left}^{i,1}(n)])$ and $f([\tilde{\mathbf{A}}_{right}^{i,1}(n)])$ of each sub-band component $\mathbf{S}_{i,1,:}$ are shown in each subgraph of Fig. 5(a). These curves indeed reveal certain distributions. It is not very meaningful to study how well they obey Rician distribution, but how to decrease the overlapped area between the two curves in each subgraph. $f([\tilde{\mathbf{A}}_{left}^{i,1}(n)])$ and $f([\tilde{\mathbf{A}}_{right}^{i,1}(n)])$ of each sub-band component $\mathbf{S}_{i,1,:}$ are shown in each subgraph of Fig. 5(b). Compared to Fig. 5(a), the function curves in most

subgraphs of Fig. 5(b) don't change much. Thus, the intrinsic envelope-distributions can be relatively accurately estimated by DoNet when $\Delta n = 1$ i.e., $\Delta t \rightarrow 0$. The estimation error will be reported in the following section. $f([\tilde{\mathbf{A}}_{100,left}^n])$ and $f([\tilde{\mathbf{A}}_{100,right}^n])$ of each sub-band component are shown in each subgraph of Fig. 5(c). Compared to Fig. 5(a), the function curves in most subgraphs of Fig. 5(c) have large changes. DoNet-1 can estimate the intrinsic envelope-distributions more accurately but has worse performance than DoNet-100. The intrinsic envelope-distributions have been optimized by DoNet-100 with the aim of higher classification accuracy. Specifically, the variance and expectation of the intrinsic envelope-distributions have been adaptively adjusted as demonstrated in **Induction 2**, which will be further verified in the following section by comparing the overlapped areas.

D. Overlapped Area Variation

Because of the data-driven mechanism, the horizontal axes of the probability density functions corresponding to DoNet-1 and DoNet-100 are not aligned with that of the intrinsic envelope. Envelope optimization cannot be evaluated via their expectations and variances.

$\mathbf{O}^{ins} \in R^{24}$ was built, where $\mathbf{O}_i^{ins}$ is the overlapped area between the two function curves in the i -th subgraph of Fig. 5(a). By the same way, $\mathbf{O}^1 \in R^{24}$ corresponding to Fig. 5(b) and $\mathbf{O}^{100} \in R^{24}$ corresponding to Fig. 5(c) were calculated respectively. $\bar{\mathbf{O}}^{ins} \in R^{24}$ represents the minimum overlap areas based on HEM. The area variations $\Delta \bar{\mathbf{O}}^{ins} = \bar{\mathbf{O}}^{ins} - \mathbf{O}^{ins}$, $\Delta \mathbf{O}^1 = \mathbf{O}^1 - \mathbf{O}^{ins}$ and $\Delta \mathbf{O}^{100} = \mathbf{O}^{100} - \mathbf{O}^{ins}$ were plotted in Fig. 6(a), namely $\bar{y}(x) = \Delta \bar{\mathbf{O}}_x^{ins}$, $y_1(x) = \Delta \mathbf{O}_x^1$ and $y_{100}(x) = \Delta \mathbf{O}_x^{100}$ in x - y coordinate.

It can be found that $y_1(x) \cong 0$. It is further verified that the amplitudes of the double-point couples can be estimated by DoNet-1. The estimation error can be reflected by $|\Delta \mathbf{O}^1|_1/24$, which equals to 1.46% with the actual values plugged in.

It can be found that $y_{100}(x) \ll 0$ when $x = 7, 10, 14$ and 17 , namely the overlapped areas in the 7th, 10th, 14th and 17th subgraphs of Fig. 5(c) decrease significantly compared to the ones of Fig. 5(a). In particular, in the sub-band of center frequency 23.125 Hz (the 17th subgraph), the overlapped area decreases by about 0.16. If the left- and right-hand MI samples are classified only by this sub-band component, the classification accuracy of a single double-point couple will be increased by 8%.

Meanwhile, $y_{100}(x) \ll \bar{y}(x)$ when $x = 7, 10, 14$ and 17 , namely the envelope distributions optimized by DoNet-100

TABLE V
CLASSIFICATION ACCURACY ON BCIC IV 2A FOR DIFFERENT L (%)

Different L	S1	S2	S3	S4	S5	S6	S7	S8	S9	Ave.
11	79.17	60.42	80.56	72.57	75.00	71.87	84.03	79.86	89.93	77.04
31	82.99	64.58	87.85	73.96	82.29	77.78	89.24	84.03	92.71	81.71
41	81.95	63.89	88.54	73.96	84.38	80.21	90.63	83.33	92.36	82.14
51	80.56	64.59	86.46	74.65	85.76	78.82	91.32	81.25	93.75	81.91
61	80.90	63.54	85.42	73.27	84.72	76.39	91.32	78.13	92.36	80.67
71	79.86	60.42	82.99	68.06	79.51	72.57	89.24	79.52	90.28	78.05
91	76.39	59.37	81.60	65.97	70.83	64.58	83.34	73.61	90.97	74.07

TABLE VI
PERFORMANCE COMPARISON OF DoNet-1, DoNet-50 AND DoNet-100 ON BCIC IV 2A (%)

Different Δn	S1	S2	S3	S4	S5	S6	S7	S8	S9	Ave.
DoNet-1	75.00	61.11	70.14	58.33	75.00	70.49	81.25	70.14	79.17	71.18
DoNet-50	77.43	63.89	76.04	69.65	79.86	71.88	89.59	76.04	86.81	76.80
DoNet-100	80.56	64.59	86.46	74.65	85.76	78.82	91.32	81.25	93.75	81.91

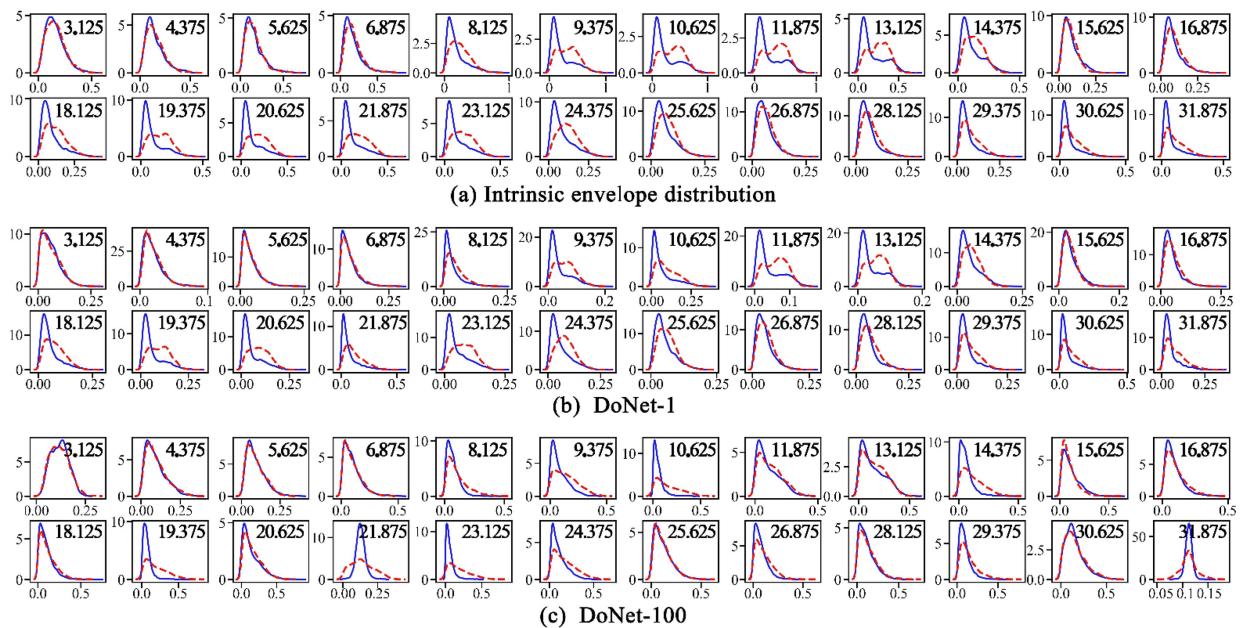


Fig. 5. Visualization result of $S_{:,1,:}$. The probability density functions corresponding to each sub-band component are shown in each subgraph. The center frequency (Hz) of the sub-band component is annotated at the right-top corner of each subgraph. (a) Probability density function of the intrinsic envelope-distribution. (b) Probability density function estimated by DoNet-1. (c) Probability density function estimated by DoNet-100.

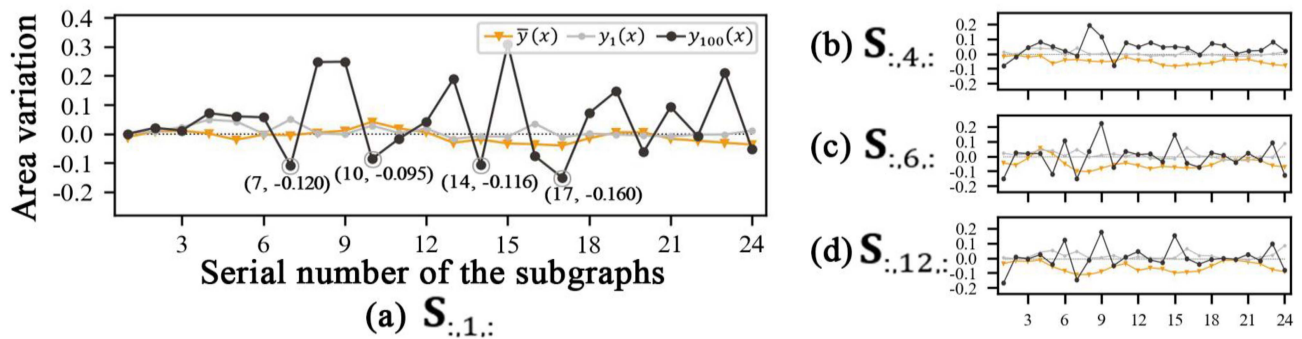


Fig. 6. Overlapped area variations.

have the smaller overlap areas than ones based on HEM. Obviously, the envelop optimization based on DoNet-100 contains extra expectation adjustments as demonstrated in **Induction 2**.

In addition, the overlapped area in some sub-bands creases significantly in Fig. 6(a). Since not all sub-band components are necessary for MI recognition, DoNet-100 can autonomously select and optimize informative sub-band components such as the 7th, 10th, 14th and 17th ones. The sub-band components whose overlapped areas increase have been excluded by ClassifierConv. This opinion can be verified indirectly by the excellent performance of DoNet-100 as shown in Table III.

The above-mentioned process was repeated on $S_{:,4,:}$ and on $S_{:,6,:}$, $S_{:,12,:}$ of the feet and tongue MI samples. Their area variations were plotted in Fig. 6(b), (c) and (d), based on which the same conclusion can be obtained.

V. DISCUSSION

A novel theoretic signal analysis-based scheme fusing model-driven and data-driven mechanisms is proposed to classify low-SNR ultra-short signals in this paper. Based on the fusion scheme, an end-to-end network instance DoNet was constructed. Specifically, the topology of DoNet is based on the proposed analytic model but the optimal model parameters are learned based on a data-driven DL network training process. The scheme has been verified by visualizing the experiment results.

A. Validation Via Visualization

It has been verified by visualization that DoNet implemented classification based on the analytic model. Corresponding to the sub-band components in the analytic model, the outputs of the FIRConv layer of a trained DoNet were analyzed and shown. The outputs of the estimation module were shown as the estimated amplitudes.

As shown in Fig. 5(a), the envelope values of the sub-band components have been shown to obey a steady distribution even if the EEG signals have very low SNR. The intrinsic envelope-distribution of the sub-band components of samples with different labels have different probability density functions. Based on the distribution discrepancy, the envelope value at each time point can be identified.

It has been verified that the amplitude of each double-point couple estimated by DoNet-1 follows the same distribution as the intrinsic envelop values (Fig. 5(b)). Therefore, each double-point couple group can be given a prediction label of satisfactory confidence. Higher classification accuracy of the samples can be obtained by a voting strategy.

More interestingly, amplitudes estimated by DoNet-100 are not from the true envelopes, but obey new distributions which can facilitate classification (Fig. 5(c)). As mentioned in **Inductions 1** and **2**, the double-point couples with larger Δt provide a free stage for the data-driven DL network to perform. Classification isn't limited by the intrinsic envelope-distribution evaluation but is based on the distributions optimized by DoNet-100 with the data-driven mechanism.

B. Combine Advantage of the Analytic Model and End-to-End Network

In fact, the analytical model can be applied as an independent classification method. However, the analytical model relies on various assumptions and approximations about real-world EEG signals. On the one hand, the estimation error is difficult to calculate accurately. It is also difficult to improve classification accuracy through error correcting even if the estimation error is obtained. On the other hand, different datasets may have different peculiarities, requiring model modifications based on expert knowledge to fit the data. In addition, pre-processing procedures relying on expert knowledge are necessary.

End-to-end deep networks can recognize minimally pre-processed EEG signals, which can be portably and widely used as a black box without expert knowledge involved in. However, the working principle of deep networks is difficult to interpret, resulting in unclear development directions. Furthermore, over-fitting problems cannot be completely solved.

In summary, analytic models may achieve higher performance than deep networks if experts try their best, but deep networks allow ordinary individuals to quickly obtain satisfactory results.

To combine the above advantages of the analytic model and end-to-end network structure, the topology and classification principle of DoNet are based on the analytic model, but the optimal parameters of DoNet are learned through a data-driven DL network training process. Meanwhile, since DoNet incorporates an adaptive spatial filter bank (SpatialConv), the pre-processing procedure in this paper only includes 1-48Hz band-pass filtering. The model-driven network structure enables solving the overfitting problem, while the data-driven parameter selection makes DoNet applicable to various MI EEG datasets. Consequently, DoNet outperforms the state-of-the-art end-to-end networks on MI EEG classification.

While DoNet has demonstrated superior performance on single-subject classification tasks compared to all state-of-the-art methods, it may not be suitable for cross-subject classification tasks. The reasons for this limitation are as follows: (1) The analytic solution presented in Section II does not cover multi-subject data domains. The alignment of domains of different subjects has not been studied. (2) The features extracted by the estimation module are assigned the predicted labels only by a simple single convolution layer, ClassConv. It is evident that ClassConv, with too few parameters and poor nonlinearity, is not suitable for cross-subject classification tasks due to the task complexity.

VI. CONCLUSION

DoNet is the first low-SNR ultra-short time-series classification model which combines the advantages of model-driven and data-driven mechanisms. The fusion of these two mechanisms is evident in two main aspects.

First, DoNet integrates data-driven noise reduction, model-driven feature extraction, and data-driven label prediction processes. The model-driven feature extraction process is formulated based on the analytical solution involving envelope calculation of double-point couples. Effective features can be obtained

with very few trainable parameters. Consequently, relatively small amount of EEG samples would be enough to fully train the data-driven noise reduction and label prediction modules. Furthermore, a large number of double-point couples can be generated from each sample, resulting in “Big Data” for MI classification. The big-data training set ensures better training of the network.

Second, the fusion of model-driven and data-driven mechanisms occurs in the feature extraction process. In fact, the envelopes of the double-point couples can be calculated using the analytical solution without any trainable parameters, which is purely model-driven. We further developed the envelope optimization theory based on independent identical distribution. By setting just a few trainable parameters in the analytical solution, we incorporate the data-driven mechanism which can extract even higher-quality features for classification.

Extensive experiments have verified that DoNet outperforms the state-of-the-art methods on 1s-long EEG classification tasks. The proposed model-driven mechanism has been verified by visualizing the experiment results on real-world EEG signals. Based on the clear model-driven mechanism and the automatic parameter learning, DoNet can be quickly deployed in various engineering applications. Conveniently, it can be modified for machine fault diagnosis, power grid stability assessment, underwater acoustic target recognition and so on. The fusion of model-driven and data-driven mechanisms is a promising research direction in the future.

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